when is the universal TM used in the proof of some important theorem/results in computational theory?

Researched 4s

A universal Turing machine (UTM) is used whenever a proof needs one machine to take another machine's description as input and execute it. Important examples include the halting problem, Rice's theorem, the time hierarchy theorem, and the invariance theorem for Kolmogorov complexity. The common ingredient is **simulation**—sometimes combined with diagonalization, sometimes with reductions, and sometimes with a bound on simulation overhead. arxiv +3

1. Halting problem: recognizability and undecidability

Consider

$$\mathrm{HALT} = \{\langle M, x\rangle : M \text{ halts on } x\}.$$

Where the UTM is used directly

To recognize HALT, run $U(\langle M\rangle, x)$ and accept if the simulation halts. On a nonhalting instance, the recognizer runs forever. Thus universal simulation gives a single recognizer for the halting behavior of all machines. The halting problem remains undecidable, including when phrased as the halting problem of a single fixed universal machine. arxiv